

Assignment 1: Introduction to ReactJS

1. Write a React program to create a **simple functional component** that displays “Hello React”.
2. Write a React program to create a **class component** that displays student information.
3. Write a React program to **create a component using JSX** that displays a heading and paragraph.
4. Write a React program to **demonstrate JSX expressions** by displaying current date and time.
5. Write a React program to **create parent and child components**.
6. Write a React program to **pass data from parent component to child component using props**.
7. Write a React program to **create multiple components and render them in App component**.
8. Write a React program to **demonstrate the use of state in a class component**.
9. Write a React program to **compare JavaScript and JSX by rendering dynamic content**.
10. Write a React program to **create a component that displays user profile using props**.
11. Write a React program to **create a counter using state in a class component**.
12. Write a React program to **convert a functional component into a class component**.
13. Write a React program to **demonstrate component lifecycle method `componentDidMount()`**.
14. Write a React program to **display a list of subjects using JSX**.
15. Write a React program to **use expressions in JSX to perform arithmetic operations**.

Assignment 2: Conditional Rendering and List

1. Write a React program to **demonstrate conditional rendering using if-else statement**.
2. Write a React program to **display login/logout button using conditional rendering**.
3. Write a React program to **use logical AND (&&) operator to render content**.
4. Write a React program to **display message if user is logged in otherwise hide it**.
5. Write a React program to **use switch case statement to display day of week**.
6. Write a React program to **prevent a component from rendering using null return**.
7. Write a React program to **display a list of student names using `map()` function**.
8. Write a React program to **render list of products with unique keys**.
9. Write a React program to **demonstrate use of React key prop in lists**.
10. Write a React program to **iterate an array and display elements using `map()`**.
11. Write a React program to **create and access refs using `useRef` hook**.
12. Write a React program to **focus an input field using React refs**.
13. Write a React program to **demonstrate event binding using `bind()` method**.
14. Write a React program to **demonstrate event binding using arrow function**.
15. Write a React program to **manage component state for a counter application**.

Assignment 3: Handling Events and Forms

1. Write a React program to **handle button click event and display alert message**.
2. Write a React program to **demonstrate event handling in React**.
3. Write a React program to **create a controlled form component**.
4. Write a React program to **create an uncontrolled form component using refs**.
5. Write a React program to **create a login form with username and password**.
6. Write a React program to **handle form submission event**.
7. Write a React program to **validate form inputs (name, email, password)**.
8. Write a React program to **set default values in input fields**.
9. Write a React program to **create a registration form using controlled components**.
10. Write a React program to **handle key events in React**.
11. Write a React program to **capture keyboard input using onKeyPress event**.
12. Write a React program to **create a form with multiple input fields and display entered data**.
13. Write a React program to **demonstrate synthetic events in React**.
14. Write a React program to **apply inline CSS styles in React component**.
15. Write a React program to **create simple animation using CSS in React component**.

Assignment 4: Routing and State Management

1. Write a React program to **create a Single Page Application using React Router**.
2. Write a React program to **configure React Router in a project**.
3. Write a React program to **navigate between Home, About and Contact pages**.
4. Write a React program to **implement dynamic routing in React Router**.
5. Write a React program to **create nested routes in React Router**.
6. Write a React program to **handle invalid URL using React Router (404 page)**.
7. Write a React program to **pass route parameters using React Router**.
8. Write a React program to **demonstrate conditional routing**.
9. Write a React program to **demonstrate state lifting between components**.
10. Write a React program to **implement global state using Context API**.
11. Write a React program to **create a Context Provider and Consumer**.
12. Write a React program to **demonstrate useState hook for counter application**.
13. Write a React program to **fetch data using useEffect hook**.
14. Write a React program to **create a custom hook**.
15. Write a React program to **focus input using useRef hook**.